SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 3 – Comparative Analysis
Weightage:	30% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2024	**Date:**	

Code of Conduct

I Mohamed Jameel A/192425057 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___________________________

Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words.	Q1
Syntax-Driven Semantic Analysis	Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.	Q3
Representing Linguistically Relevant Concepts	Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.	Q4
Information Retrieval & Syntax-	Analyze large-scale parsing strategies, evaluate performance	Q5

Driven Semantic Analysis	trade-offs, and justify efficient approaches for processing massive linguistic datasets.	

Annexure A – Comparative Analysis

1. A conversational AI system must identify grammatical relationships in user queries containing long-distance dependencies. The developers are comparing constituency parsing and dependency parsing.

Analyze how the two representations differ in capturing hierarchical and word-to-word relationships, and justify which approach is more suitable for extracting meaningful relationships from conversational text.

2. An NLP system processes partially typed search queries such as “best hotels near…” and “book a flight from Chennai to…”. The system currently uses conventional top-down parsing, while the developers are considering Earley parsing.

Analyze how the two approaches behave when input is incomplete or ambiguous, and justify which parsing strategy is more appropriate for interactive NLP applications.

3. A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. The developers are comparing CFG, PCFG, and neural constituency parsers.

Analyze how each approach represents and resolves syntactic ambiguity, and justify which approach would provide the best balance between interpretability and practical accuracy.

4. A multilingual NLP system must process sentences in which grammatical information depends on number, gender, tense, and person. The developers are considering feature structures and traditional context-free grammar rules.

Analyze how feature structures extend basic grammar representations and justify their usefulness for enforcing grammatical constraints in multilingual systems.

5. A voice assistant must interpret commands containing verbs with different argument requirements, such as “give the book to John,” “sleep,” and “put the file on the desk.” The developers are considering subcategorization frames.

Analyze how subcategorization information helps determine valid sentence structures and justify its importance in semantic and syntactic analysis.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Scope of Items	Comparison Depth	Use of Evidence	Critical Thinking	Clarity of Presentation	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Q1 — Constituency Parsing vs Dependency Parsing

A conversational AI system needs to identify grammatical relationships in queries containing **long-distance dependencies**. The two approaches being compared are **constituency parsing** and **dependency parsing**.

Comparison

Aspect	**Constituency Parsing**	**Dependency Parsing**
Basic representation	Represents a sentence as **hierarchical phrases/constituents**	Represents **word-to-word dependencies**
Structure	Builds a tree of phrases such as NP, VP	Builds relationships between individual words
Main focus	Shows how words form larger grammatical units	Shows which words depend on or modify other words
Long-distance relationships	Can represent them through hierarchical structure, but relationships may require traversing the tree	Direct dependency links make relationships between words easier to identify
Semantic extraction	Useful for identifying phrase structure	Particularly useful for extracting **who did what to whom**
Conversational queries	Useful when phrase structure is important	More direct for extracting relationships from flexible conversational language

Example

Consider:

“Which book did the student say the teacher recommended?”

A constituency parser represents the nested phrase structure:

```
S
├── NP → Which book
└── VP
    ├── V → did
    └── ...
        └── recommended → which book
```

A dependency parser focuses on the relationships between words, making the relationship between **recommended** and **book** explicit even though they are separated by other words.

Advantages

Constituency parsing

- Provides a clear hierarchical representation.
- Useful for identifying phrases and nested structures.
- Helps when the internal structure of a sentence is important.

Dependency parsing

- Gives direct word-to-word relationships.
- Makes semantic relationships easier to extract.
- Is well suited to identifying relations such as **subject–verb–object**, especially in conversational text.

Best Choice

For this **conversational AI system**, **dependency parsing is more suitable for extracting meaningful relationships**. The main requirement is to identify relationships between words despite long-distance dependencies, and dependency links directly represent those relationships. However, constituency parsing remains useful when the system needs detailed **phrase-level and hierarchical structure**.

Conclusion

Dependency parsing is the better primary representation for this scenario, while constituency parsing can complement it when detailed phrase structure is required. This conclusion follows from the difference specified in the question: constituency parsing emphasizes **hierarchical structure**, whereas dependency parsing emphasizes **word-to-word relationships**.

Q2 — Top-Down Parsing vs Earley Parsing

The NLP system processes incomplete search queries such as **“best hotels near…”** and **“book a flight from Chennai to…”**. It currently uses conventional top-down parsing and is considering Earley parsing.

Comparison

Aspect	**Top-Down Parsing**	**Earley Parsing**
Basic approach	Starts from the start symbol and predicts grammar rules	Builds possible parses while scanning the input
Incomplete input	Can struggle because the expected structure may not be complete	**Better suited** to partial input
Ambiguity	May require significant backtracking	Maintains multiple possible parse states
Backtracking	Can become inefficient	Avoids much of the repeated backtracking of conventional top-down parsing
Complex queries	Efficiency decreases as alternatives increase	Handles complex and ambiguous structures more effectively
Interactive NLP	Less suitable	**More suitable**

Example

For:

best hotels near...

the input is incomplete. A top-down parser may attempt to complete the expected grammar structure and encounter missing constituents.

Earley parsing can maintain the **partial parse state** and continue when additional words arrive.

Similarly:

book a flight from Chennai to...

can be processed as partial input until the destination is supplied.

Ambiguity

Suppose a query can have more than one valid syntactic interpretation. Top-down parsing may explore alternatives through backtracking, increasing processing time.

Earley parsing can maintain different possible parse structures while processing the input, making it more suitable for ambiguous interactive queries.

Best Choice

Earley parsing is the better choice for interactive NLP applications.

The reason is that the system specifically needs to handle:

- incomplete user input,
- ambiguous queries,
- and real-time interaction.

These are exactly the conditions identified in the scenario.

Conclusion

While top-down parsing is simpler for well-formed complete sentences, **Earley parsing provides better flexibility for partially typed and ambiguous search queries**. Therefore, replacing the conventional top-down parser with Earley parsing would make the interactive search system more robust.

Q3 — CFG vs PCFG vs Neural Constituency Parsing

The grammar-checking system must distinguish between **multiple interpretations of structurally ambiguous sentences**. The three approaches being compared are CFG, PCFG, and neural constituency parsers.

Comparison

Aspect	**CFG**	**PCFG**	**Neural Constituency Parser**
Representation	Phrase structure using grammar rules	Phrase structure + probability for each rule	Phrase structure learned from data
Ambiguity	Can produce multiple valid parse trees	Ranks possible parses using probabilities	Uses learned contextual patterns to select likely parses
Interpretability	**Very high**	**High**	Lower
Context handling	Limited	Better through probabilities	**Strong**
Data requirement	Grammar rules	Grammar + probabilities	Requires training data/model
Accuracy on natural language	Limited	Better than basic CFG	**Generally strongest in practical use**
Explainability	Easy to understand	Relatively explainable	More difficult to explain

Example

For an ambiguous sentence with two possible structures:

CFG:

↓

Parse 1

Parse 2

CFG can represent both possibilities, but it does not inherently know which interpretation is more likely.

With PCFG:

Parse 1 → 0.75

Parse 2 → 0.25

Choose Parse 1

Thus, PCFG uses probabilities to resolve ambiguity.

A neural constituency parser can use **context learned from language data** to select the most appropriate structure rather than relying only on manually specified grammar probabilities.

Analysis

CFG has the strongest interpretability because its decisions can be directly traced to explicit grammar rules. However, its ability to resolve natural-language ambiguity is limited.

PCFG improves on CFG by assigning probabilities to alternative structures. It therefore provides a useful balance between explicit grammar and ambiguity resolution.

Neural constituency parsing can capture richer contextual patterns and is better suited to practical natural-language ambiguity, but its decisions are less transparent.

Best Choice

For this grammar-checking system, **PCFG provides the best balance between interpretability and practical accuracy**.

CFG

→ Highly interpretable

→ Weak ambiguity resolution

PCFG

→ Interpretable

→ Probabilistic ambiguity resolution

→ Good balance

Neural Parser

→ Strong contextual accuracy

→ Lower interpretability

Conclusion

PCFG is the most balanced choice when both explainability and practical ambiguity resolution are important. A neural constituency parser would be preferable if maximum practical parsing accuracy were prioritized over interpretability.

Q4 — Feature Structures vs Traditional CFG

The multilingual NLP system must handle grammatical information such as **number, gender, tense, and person**. The two approaches being compared are traditional CFG rules and feature structures.

Comparison

Aspect	**Traditional CFG**	**Feature Structures**
Representation	Uses grammar production rules	Adds grammatical features to linguistic structures
Number	Requires separate rules for different forms	Can represent singular/plural as a feature
Gender	Difficult to encode efficiently	Can represent masculine/feminine/etc.
Tense	Usually needs separate productions	Can represent tense as a feature
Person	Requires additional grammar rules	Can directly represent person
Agreement	Limited	**Explicitly enforces agreement**
Multilingual use	Grammar rules can become large	**More flexible and scalable**
Grammar complexity	Increases as constraints increase	Reduces repeated rules by sharing features

Example

Consider:

The boy runs.

The system can represent:

Subject:

Number = Singular

Person = 3

Verb:

Number = Singular

Person = 3

The features must agree.

For:

The boys run.

the representation becomes:

Subject:

Number = Plural

Verb:

Number = Plural

A traditional CFG may require separate rules such as:

S → NP_s VP_s

S → NP_p VP_p

As more features such as gender, tense, and person are introduced, the number of grammar rules can grow significantly.

Why Feature Structures Are Better

Feature structures allow grammatical constraints to be represented **explicitly as attributes and values**:

[Number: plural

Gender: feminine

Person: 3

Tense: present]

They can therefore enforce compatibility between different parts of a sentence instead of creating a separate CFG rule for every possible combination.

This is particularly useful in **multilingual systems**, where grammatical features and agreement requirements can vary across languages.

Best Choice

Feature structures are more suitable for the multilingual system.

They provide:

- explicit grammatical constraints,
- better agreement handling,
- compact linguistic representation,
- greater flexibility when multiple grammatical features interact.

Conclusion

Traditional CFG is useful for representing basic syntactic structure, but **feature structures extend CFG-style representations with grammatical attributes such as number, gender, tense, and person**. Therefore, they are better suited for enforcing grammatical constraints in a multilingual NLP system.

Q5 — Subcategorization Frames

The voice assistant must interpret verbs with different **argument requirements**, such as:

- "give the book to John"
- "sleep"
- "put the file on the desk"

1. What Subcategorization Frames Do

A subcategorization frame specifies **what kinds of complements a verb requires** to form a valid sentence.

Verb	Frame	Example
give	Verb + NP + PP	Give **the book to John**
sleep	Verb only	Sleep
put	Verb + NP + PP	Put **the file on the desk**

So the system can determine which arguments are expected for each verb.

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2. Why It Matters for Parsing

Without subcategorization information, the parser may not know whether a sentence is complete or what role each phrase plays.

For example:

Put the file...

is incomplete because **put** normally requires a location:

Put the file on the desk.

Similarly:

Sleep the book...

would violate the expected frame for **sleep**, because sleep does not normally take an object.

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3. Role in Semantic Analysis

Subcategorization frames help connect verbs with their semantic participants.

For:

Give the book to John.

the system can derive:

Action → Give

Object → Book

Recipient → John

For:

Put the file on the desk.

Action → Put

Object → File

Location → Desk

This allows the voice assistant to convert a sentence into a structured meaning rather than simply identifying individual words.

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4. Importance in the Voice Assistant

Subcategorization frames help the system:

- identify **required arguments**
- detect **incomplete commands**
- reject grammatically invalid structures
- assign semantic roles to arguments
- improve command interpretation

Conclusion

Subcategorization frames provide the parser with information about **which arguments a verb requires and how those arguments participate in the action**. They are therefore important for both **syntactic parsing and semantic interpretation**, especially in a voice assistant where commands must be converted into structured actions.